

Lesson 3 · Homework

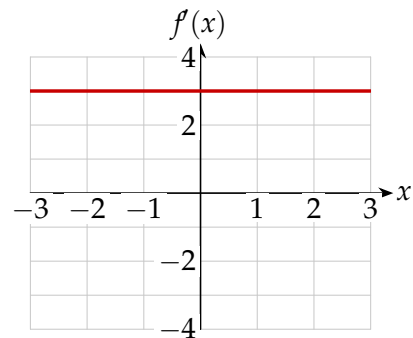
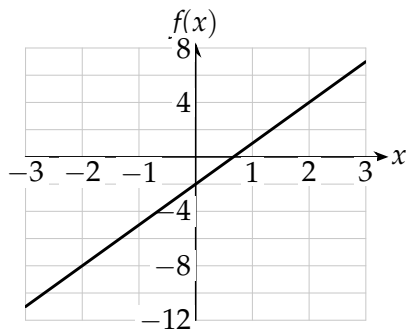
NCEA Level 2 · AS 91262 · Sketching gradient functions

Time: about 60 minutes

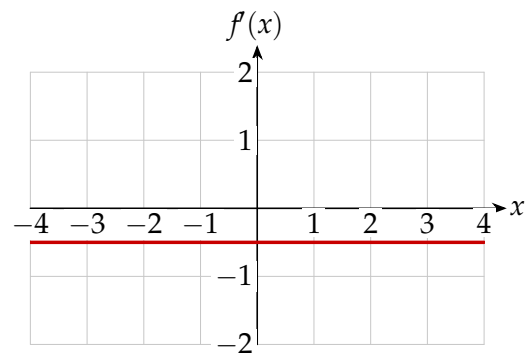
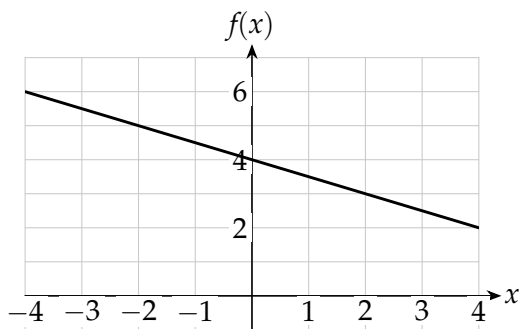
Answer version · tutor's working, not an official mark scheme

Sketch on the empty axes. Line up every key x -value with a ruler, and label turning points. [A] Achieved, [M] Merit, [E] Excellence.

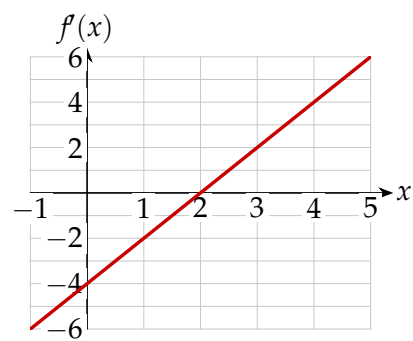
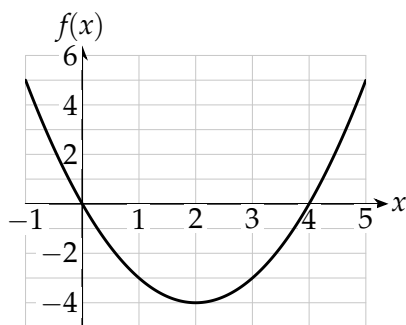
1. The graph of $f(x) = 3x - 2$ is shown. Sketch $f'(x)$. [A]



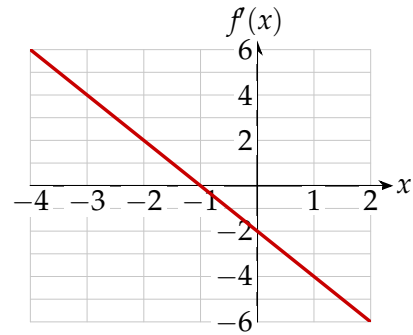
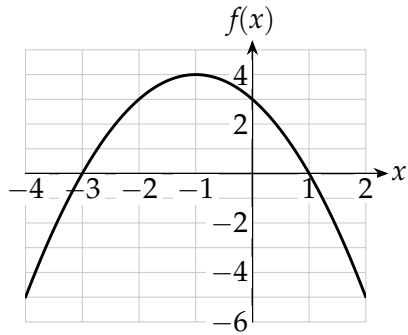
2. The graph of $f(x) = 4 - \frac{1}{2}x$ is shown. Sketch $f'(x)$. [A]



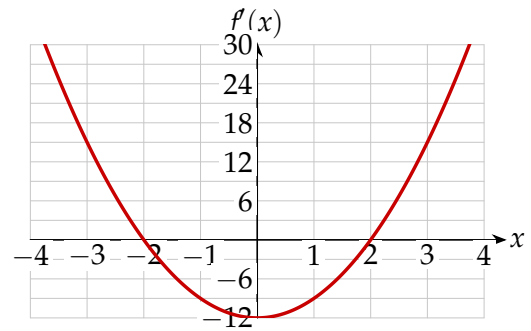
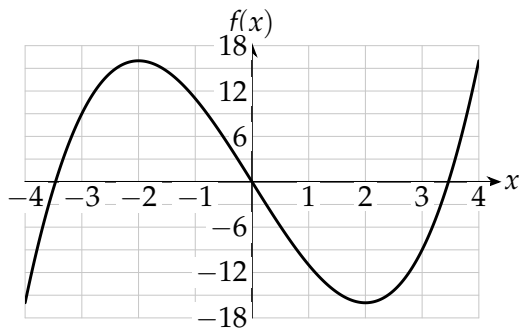
3. The graph of $f(x) = x^2 - 4x$ is shown. Sketch $f'(x)$. [A]



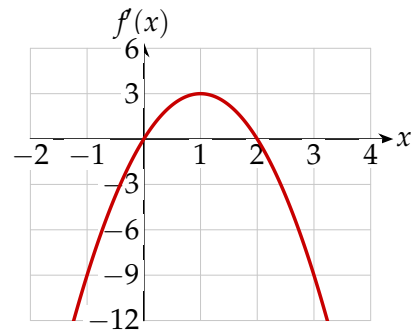
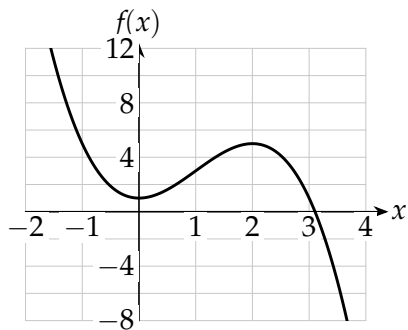
4. The graph of $f(x) = 3 - 2x - x^2$ is shown. Sketch $f'(x)$. [A]



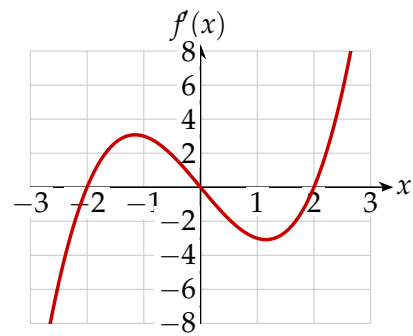
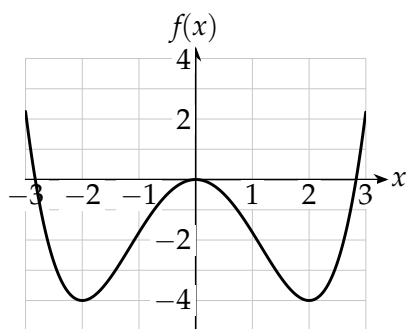
5. The graph of $f(x) = x^3 - 12x$ is shown. Sketch $f'(x)$. [A]



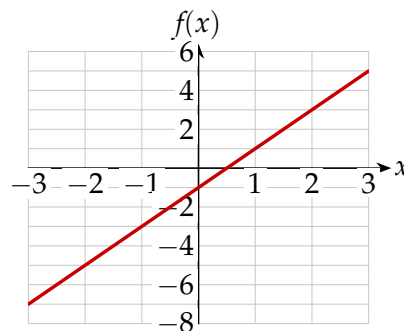
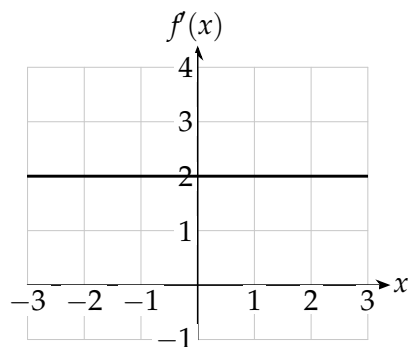
6. The graph of $f(x) = -x^3 + 3x^2 + 1$ is shown. Sketch $f'(x)$. [A]



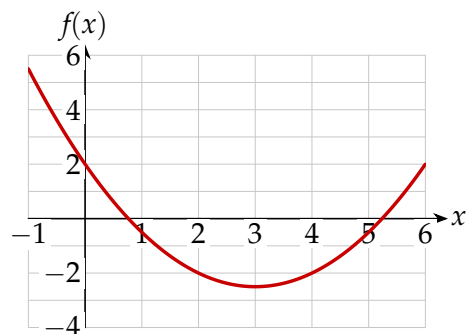
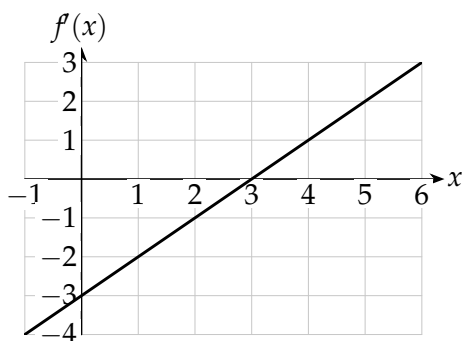
7. The graph of $f(x) = \frac{1}{4}x^4 - 2x^2$ is shown. Sketch $f'(x)$. [M]



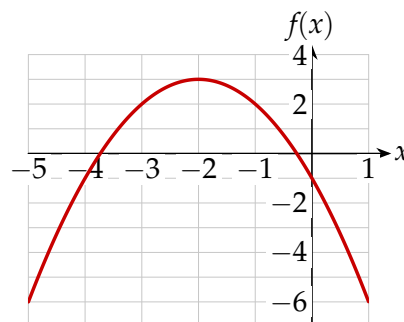
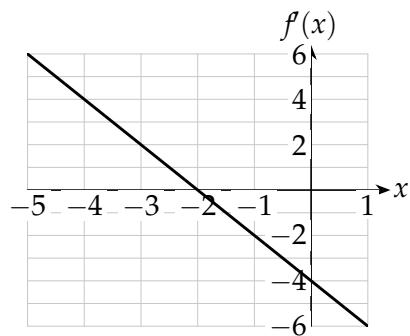
8. The graph of $y = f'(x)$ is shown. Sketch a possible graph of $f(x)$. [A]



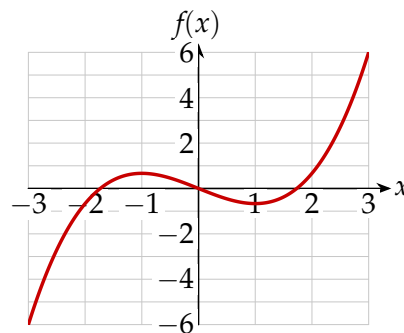
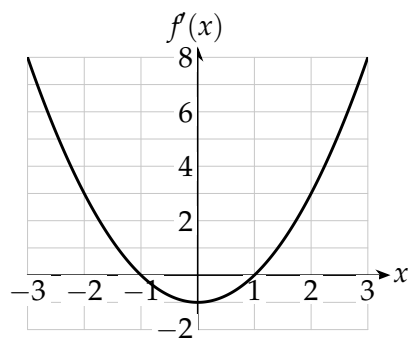
9. The graph of $y = f'(x)$ is shown. Sketch a possible graph of $f(x)$. [A]



10. The graph of $y = f'(x)$ is shown. Sketch a possible graph of $f(x)$. [A]

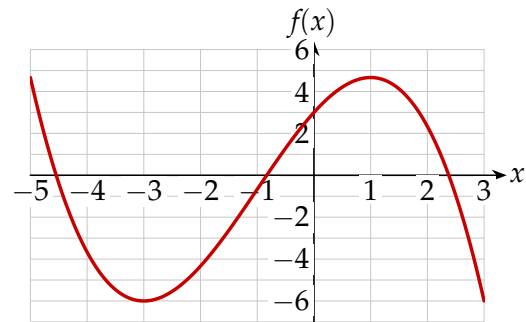
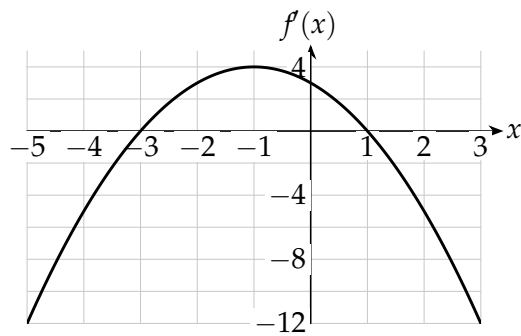


11. The graph of $y = f'(x)$ is shown. Sketch a possible graph of $f(x)$. [M]



12. The graph of $y = f'(x)$ is shown. Sketch a possible graph of $f(x)$ and label its maximum and minimum.

[M]



13. Tama says: “Where the graph of $f'(x)$ has its highest point, the graph of $f(x)$ has a maximum.” Is Tama right? Explain, using $f(x) = x^3 - 3x$ (so $f'(x) = 3x^2 - 3$) or an example of your own.

[M]

No. A maximum of f is where $f'(x) = 0$ and f' changes from $+$ to $-$.

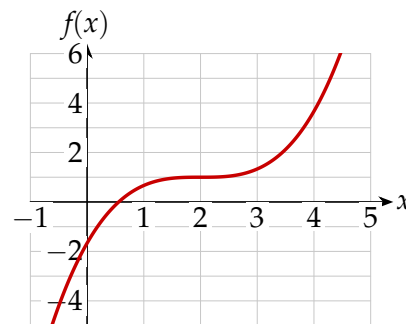
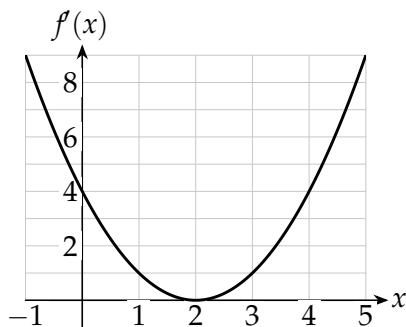
A turning point of f' is where f is steepest (a point of inflection), not a turning point of f .

E.g. $f'(x) = 3x^2 - 3$ has its lowest point at $x = 0$, where f is steepest downhill;

the maximum of f is at $x = -1$, where $f'(-1) = 0$.

14. **Extension** The graph of $f'(x) = (x - 2)^2$ touches the x -axis at $x = 2$ but does not cross it. Sketch a possible graph of $f(x)$, and describe what happens to f at $x = 2$.

[E]



$f'(x) \geq 0$ everywhere, so f is always increasing.

At $x = 2$, $f' = 0$ but does not change sign: f is flat there (a stationary point of inflection),

not a maximum or a minimum. One possible $f(x) = \frac{1}{3}(x - 2)^3 + 1$.

Optional extra: StudyTime Level 2 Calculus Guide, “Sketching Gradient Functions” from page 19.